Case study 3: Bird box

National 5 · Practical Woodworking · Answers and marking guidance

Award one mark per correct point unless stated. Other valid answers should be credited. Sketches can gain full marks if they show enough detail.

1 (a) Identify an appropriate wood for the bird box. (1 mark)

- Cedar
- Larch
- Oak

Any 1 point.

1 (b) Identify a property that makes this wood suitable. (1 mark)

- Natural oils / naturally durable, so it resists rot outdoors

2 State a suitable adhesive for the bird box and give a reason. (2 marks)

- Exterior / waterproof PVA, or polyurethane / resin glue
- It is waterproof, so the joints won't fail in wet weather

3 Describe how to stop the wood splintering where the bit comes through. (1 mark)

- Clamp a scrap piece of wood behind the front and drill into it
- Drill from both sides, meeting in the middle

Any 1 point.

4 Describe three things the manufacturer could do to make the bird box more environmentally friendly. (3 marks)

- Use FSC-certified, reclaimed or local timber
- Plan the cutting list to reduce waste; use offcuts
- Use water-based/eco finishes and glues, or leave unfinished (cedar)
- Design so it can be repaired; use recyclable packaging

Any 3 points.

4 Give one reason why a finish is not applied inside the box. (2 marks)

- Finishes can give off fumes or be harmful
- So the chicks / nesting birds are not harmed

Total: 10 marks